Health Expenditures and Poverty -

A panel data analysis of the effect of out-of-pocket health expenditures on poverty across 143 countries from 2000 to 2019

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Abstract

The objective of this research is to examine the relationship between out-of-pocket health expenditures as a proportion of total health expenditure and poverty levels. Data from 143 countries over the period of 2000 to 2019 is used, employing four different proxies for poverty and three distinct poverty lines. Based on the empirical analysis presented, it can be concluded that there is a significant positive association between out-of-pocket health expenditures as a share of total health expenditures and poverty, considering poverty lines of 2.15 USD and 3.65 USD. Since this study uses non-experimental data, control variables are included in the regression analysis aiming to reduce omitted variable bias. Given the specific sample utilized in the study and keeping everything else constant, a one percent increase in the share of out-of-pocket health expenditures as a proportion of total health expenditure is expected to increase, on average, the poverty headcount by 0.1138 (0.1501) percent when using a poverty line of 2.15 (3.65) USD per day. These findings hold statistical significance at the one percent confidence level.

Introduction

After the World Bank released its poverty estimates in 2010, it was made clear that one of the essential human needs is still not guaranteed. It was estimated that 15,7% of the world population lives under the standard $1.90 per day poverty line (World Bank, 2010). For this group, access to healthcare is not guaranteed, or simply too expensive. The approaches to fighting poverty have been improving over time. Leading institutions and governments have invested large amounts of time and money to reduce poverty numbers, with reasonable success (WHO, 2020). But still, many low-income countries are lagging far behind.

In the same year, the World Health Organization released their Universal Health Coverage Agenda (WHO, 2010). A part of this agenda focused on improving financial protection against medical expenditures for households. In a substantial number of poor countries, the public share of total healthcare spending is much lower than in higher-income countries. Consequently, the share of out-of-pocket healthcare expenditures is a lot higher, often above 50% (WHO, 2010). In poorer countries, these healthcare expenditures can fill up a large part of a household's total spending. Therefore, the WHO agenda partly focused on reducing the financial impact of out-of-pocket healthcare expenditures on households. Out-of-pocket healthcare expenditures are medical expenses that are be reimbursed by insurance providers.

The World Health Organization also noted that out of the entire world population, 1.77% was pushed below the $1.90 per day poverty line by out-of-pocket healthcare expenditures (WHO, 2010). This might therefore suggest that an increase in out-of-pocket healthcare expenditures has the potential to cause worse economic circumstances for households, resulting in poverty.

The WHO base their results on surveys of households in each country. For each country, a representative sample of the population is surveyed. For this sample, spending, income patterns and financial situations are examined. The results found in this sample are then generalized for the entire population and published in the World Healthcare Statistics (WHO, 2021).

While the surveys of the World Health Organization are conducted in every country, there has been only limited research trying to relate poverty and out-of-pocket health expenditure. Past literature mostly focused on nationwide survey data intending to show that

there exists a significant relationship between the two aforementioned measures in developing countries, such as Ghana and Bangladesh (Novignon, Mussa, & Chiwaula, 2012). Our research, therefore, adds to the existing literature as we use macro-level aggregated data in a linear regression model instead of survey-based data.

More precisely, we include multi-variate data for countries around the world, over a time span of multiple decades, controlling for macro-economic and socio-demographic factors that possibly affect poverty. We are then able to examine how the prevalence of poverty in countries is related to their out-of-pocket medical expenditures, whilst controlling for other regressors related to poverty. Additionally, we can then look at a substantial number of countries at a time and do not have to limit our research to one country.

Furthermore, out-of-pocket expenditures on healthcare have reportedly been a major reason for impoverishment at the household level. Wagstaff, & Doorslaer (2003) and Brown et al. (2014) discovered that poorer households tend to avoid or delay healthcare treatments as a consequence, especially if they involve out-of-pocket spending. Therefore, it is relevant for society as a whole to examine how out-of-pocket healthcare expenditures are related to poverty.

As this relationship between out-of-pocket healthcare expenditures and poverty has not been examined on an aggregate level, it could result in new policy motives. Our analysis cannot be used to infer any causal effect of out-of-pocket medical expenditures on poverty. But, if a general association between the two is established, a substantial basis to invest in further research examining a possible causal relation is created. It could very well be that a part of the possible positive association is consistent with a causal relation. Besides, it yields a strong argument to further examine the levelling effect and effectiveness of policies reducing out-of-pocket healthcare expenditures. We will thus examine the following research question:

To what extent are out-of-pocket and government expenditures for healthcare associated with poverty in countries across the world from 2000 to 2019?

Theoretical framework

Past research has tried to examine how health expenditures affected poverty. But the research has been quite limited. Xu et al. (2003), Hamid et al. (2014) and van Minh et al. (2013) found significant positive relations between out-of-pocket health expenditures and poverty in some specific underdeveloped regions and countries. Although a significant relation was found, most of these papers relied on country-based surveys or small data samples which made implications very limited. Most papers were solely based on respondents from low-income countries such as Vietnam, India, China and Bangladesh. Xu et al. (2003) made a more extensive analysis based on 65 countries.

The first paper that examined the relationship between out-of-pocket healthcare expenditures and poverty using macroeconomic data containing a larger set of countries was that of Sirag and Nor (2021), which used a dynamic panel threshold model for cross-country analysis. This model, proposed by Seo and Shin (2016), allows for the endogeneity of the threshold regressor out-of-pocket healthcare expenditures by introducing a first-differenced GMM estimator for each independent variable. This paper found that, to some threshold, out-of-pocket healthcare expenditures are significantly associated with poverty. The limitation of this paper was that it only included a single poverty line in its regression model, which captured extreme poverty and only used GDP per capita as control variable.

Regarding poverty lines, we will follow the definition of the World Bank (2022). It defines extreme poverty as having less than 2.15 USD of income per day. Estimations by the World Bank considered 648 million people to live under this threshold in 2019. Furthermore, the World Bank decided to introduce a poverty line for lower-middle-income countries set to 3.65 USD of income per day and a poverty threshold for upper-middle-income countries set to 6.85 USD of income per day. Considering Sirag and Nor's (2021) recommendation for further research, we will alter the poverty line in accordance with the decision of the World Bank (2022) to raise it from 1.90 USD to 2.15 USD. Additionally, to expand already existing research, we also include the poverty lines of lower-middle-income and upper-middle-income countries.

In the past, some research has been performed analyzing the effects of out-of-pocket healthcare expenditures on poverty by means of survey analysis. In this research paper, however, we will focus on the analysis of the aforementioned relationship via country-level

variables in a linear regression model, where we control for the endogeneity of the out-of-pocket expenditures for healthcare in our regression. To do so, we will include a set of relevant control variables based on earlier research. Besides, to control the unobservable differences between countries, we will add dummy intercepts for each country in our model, besides our regressor of interest. To control for unobservable differences over time, we will also include dummy intercepts for each yearly observation.

To ensure that we do not measure a distorted relationship, we shall carefully focus on how the existing literature describes significant determinants for poverty. We will therefore further extrapolate relevant control variables. A substantial number of empirical studies have been carried out that focus on the dynamic relationship between macroeconomic variables and different measures of poverty as suggested by Jefferson and Kim (2012).

Poverty was found significantly negatively related to general macroeconomic variables which determine the overall economic state of a country. The Real Gross Domestic Product (GDP) per capita is a well-known macroeconomic measure which is related to poverty, as suggested by Cerra et al. (2021). The real GDP represents the value of total economic production in a country, corrected for inflation. It, therefore, estimates the real aggregate national income over time. It is therefore plausible to assume that real GDP per capita, to some extent, affects poverty in countries. A more extended description of the relationship between GDP and poverty is given by Powers (1995).

Furthermore, the national unemployment rate is one of the most important determinants of poverty as suggested by Blank (2000). Its relation to poverty is theoretically straightforward: if unemployment increases, the average income of the poor will decrease, and poverty is more likely to occur in lower-income groups. It was found by Saunders (2002) and DeFina (2004), that the relation between unemployment rate and poverty is dependent on how poverty rates are measured. A higher baseline poverty measure caused a reduction in the measured impact of the unemployment rate on poverty. This motivates the inclusion of multiple poverty lines in our analysis. Saunders (2002) examined the relationship between unemployment rates on different poverty measures on aggregated macro-economic data and DeFina (2004) examined this relation using aggregated US data. Both papers concluded that the unemployment rate has a significant negative effect on the poverty headcount in a country.

For some macroeconomic variables, a significant relation with poverty is not unitarily attached on a macro level. The effect of inflation on poverty is often found to be statistically insignificant (Jefferson & Kim, 2012). But it was also shown by Blank (1986) that inflation has a mild increasing effect on the poverty rate over a large dataset, or even a strong effect, as suggested by Balke and Slottje (1993).

It has been shown, by studies like those of Piachaud (2002), Rupasingha and Goetz (2007) and Achia et al. (2010), that different social conditions, labor environments and demographic properties of countries are significant determinants of poverty in different types of countries.

Therefore, we choose nationwide measured indicators as control variables used as proxy variables for social conditions in a country, which are possibly associated with poverty. We consider the rural population ratio as Polloni-Silva et al. (2021) have done when analyzing the effect of financial inclusion on diminishing poverty and income inequality. A critical review of the current literature on the association between rurality and poverty was conducted by Weber and Jensen (2004). This paper underlines a possible significant association between rurality and poverty but warns of methodological concerns in the research that was conducted.

Furthermore, as Jamil et al. (2020) showed in their research, the inclusion of female workers and their share of the total workforce are highly relevant to mitigate poverty. This also intuitively makes sense, as females represent roughly half of the population and therefore are important to consider when looking at the financial situations of individuals. Rahman (2013) conducted a logistic regression analysis to identify which household characteristics affect measures of poverty for a household, based on survey data in Bangladesh. In this study, it was found that the risk of being poor is higher for households where female members and workers predominate. Although this yields a different perspective on female workforce inclusion, it further motivates the usage of female workers as a share of the total workforce as a control variable.

Data

In order to be able to perform the linear regression, we use panel data from countries all over the world. The variables of interest are studied in the period from 2000 to 2019, leaving out the entire Covid-19 pandemic. This is done to be able to study the effects of out-of-pocket healthcare spending on poverty without the additional effects of the global pandemic on that relation. In total, 1449 observations from 143 countries are obtained for the regression analysis using only those observations where data for all variables of interest are present. Countries with only one observation are excluded as these countries do not contribute to the regression when using fixed effects. The list of used countries including their income group according to the World Bank and their unique ISO 3166-1 alpha-3 codes are presented in Table A1 in Appendix A.

The main dependent variable that we are going to examine is poverty. In academic literature, poverty headcount, poverty gap, squared poverty gap and the Watts index are usually used as proxies for poverty in a country. The poverty headcount ratio is the percentage of the population which is living in households with consumption or income below the poverty line at 2017 international prices. The poverty gap is the mean shortfall in income or consumption from the poverty line a day expressed as a percentage of the poverty line. It not only measures the incidence of poverty in a specific region or country but also includes the depth of poverty. The poverty gap squared is sometimes also called the poverty severity index as it increases the weight the farther away a person's income falls below the poverty threshold by squaring the terms. The Watts index represents the mean of the population of poverty gaps measured by the log of the ratio of the poverty line to income.

All the variables that we use in the regression analysis and the sources the data are obtained from are summarized in Table 1. To keep the variable names of the dependent variables, the independent variable of interest and the control variables in the following tables as short as possible, we refer to the variables using the names introduced in Table 1 throughout this paper. To indicate which poverty line is referred to when presenting poverty indicators, the poverty line will be indicated after the name of the poverty measure with 2.15 USD, 3.65 USD or 6.85 USD.

Table 1 – Data sources overview

Variable	Data Source	Measuring unit and scale
Poverty headcount ratio	Poverty and Inequality Platform (The World Bank, 2023)	Percentage between 0 and 100
Poverty gap	Poverty and Inequality Platform (The World Bank, 2023)	Percentage between 0 and 100
Poverty gap squared	Poverty and Inequality Platform (The World Bank, 2023)	Percentage between 0 and 100
Watts index	Poverty and Inequality Platform (The World Bank, 2023)	Percentage between 0 and 100
Share of out-of-pocket expenditure	The World Bank (2022)	Percentage of the out-of-pocket health expenditures as share of total health expenditure between 0 and 100
Real GDP per capita	World Development Indicators (The World Bank, 2023)	Natural logarithm of the real value
Unemployment rate	International Labor Association (2023)	Percentage of the unemployed as a share of the total labor force between 0 and 100
Share of rural population	The World Bank (2023)	Percentage of the rural population as a share of the total population between 0 and 100
Share of female workforce participation	World Development Indicators (The World Bank, 2023)	Percentage of the female workforce as a share of the total labor force between 0 and 100

Notes: This table is representing the variables used in the regression analysis. The left column shows the variable names, the middle column includes the source used to retrieve the data and the year the database was last updated. On the right-most column the measuring unit and scale of each variable is indicated. The exact definition of the poverty measures is explained in the main text.

Table 2 presents the descriptive statistics for the poverty headcount ratio using different poverty lines, the main independent variable of interest and the control variables. Included in Appendix B, Table B1 shows additional descriptive statistics for all four poverty measures used in the regression analysis.

Table 2 – Descriptive statistics

Variable	Obs.	Mean	St. Dev.	Min.	Max.
Poverty headcount ratio 2.15 USD	1449	7.41	14.77	0.0000	91.47
Poverty headcount ratio 3.65 USD	1449	15.43	23.09	0.0000	97.76
Poverty headcount ratio 6.85 USD	1449	29.54	31.39	0.0000	99.70
Share of out-of-pocket expenditure	1449	31.49	16.73	2.17	84.79
Real GDP per capita	1449	24059.38	19961.66	731.99	120647.80
Unemployment rate	1449	8.00	5.35	0.25	37.32
Share of rural population	1449	35.62	18.68	1.96	90.38
Share of female workforce participation	1449	43.32	6.32	14.58	60.58

Notes: this table is representing the descriptive statistics which characterize this dataset. The left column shows the variables under analysis. Furthermore, the number of observations, the sample mean, the sample standard deviation, the minimum values and the maximum values recorded in the sample are included. All variables are measured in accordance with the units presented in the data section.

To get a first impression of the dataset, Figure 1 plots the share of out-of-pocket health expenditures against the poverty headcount ratio using the 2.15 USD poverty line. Even though all of the observations obtained are shown in the scatterplot and country- and time-specific adjustments have not been made, it is possible to observe a positive relationship between out-of-pocket health expenditure and poverty. However, this is only a preliminary conclusion and is tested using linear regression.

Furthermore, Figures C1 to C4 in Appendix C show the same scatterplot in terms of the relation between out-of-pocket health expenditures and poverty, but each only contains observations from one income group. When looking at the scatterplots, the suspicion that the relation between out-of-pocket health expenditure and poverty is affected by the income status of the country arises. Therefore, we also separate between income groups when performing the regression analysis. It is important to mention that this is also only a rough first impression.

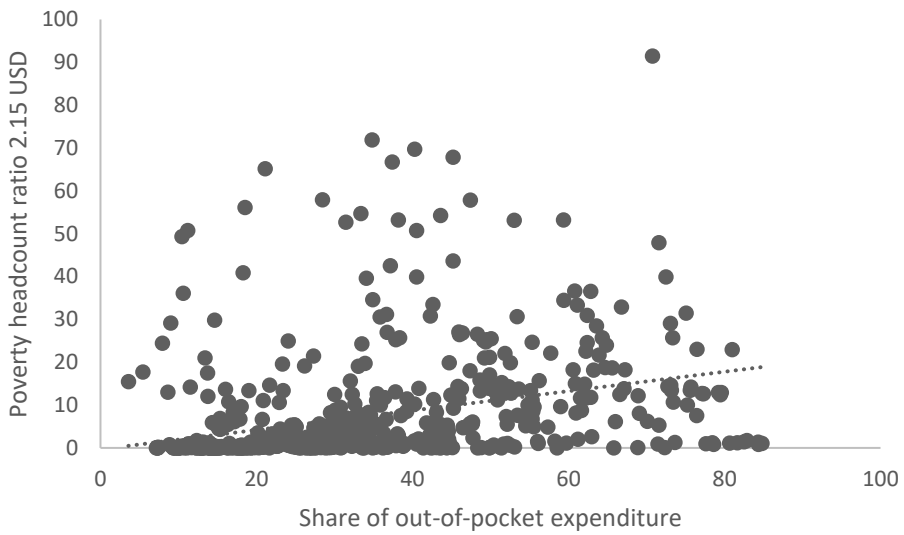

Figure 1 – Scatterplot of share of out-of-pocket health expenditures against the poverty headcount ratio at the 2.15 USD poverty line

Notes: The scatterplot shows the relationship between the out-of-pocket health expenditures as share of total health expenditures and the poverty headcount ratio using a poverty line of 2.15 USD per day. Both measurements are given as percentages between 0 and 100. Each point on the graph represents one observation of the dataset.

Table 3 summarizes how many observations stem from which income group and how many countries are contained in each income group. From Table 3 it becomes clear that more observations are missing in low-income countries than in high-income countries. This suggests that the unbalanced panel data that is used in the regression analysis could lead to biased estimates. A reason for this is that it is highly likely that the missing observations are not random, but more are missing among countries with lower income.

Table 3 – Division into income groups

Variable	Observations	Countries	Years
High income	654	39	20
Upper-middle income	450	38	20
Lower-middle income	273	46	20
Low income	72	20	19
Total	1449	143	20

Notes: The left-most column indicates the income group as classified by the World Bank. The remaining columns show the number of observations, countries and years, respectively in each income group.

Methodology

In order to analyze the panel data that we have collected, we use a fixed effects model. In the fixed effects model, we include dummy intercepts for every country such that countries can differ in the level they are attaining but have the same slope coefficient in the end. To control for time-varying factors across countries year fixed effects are included in our model. This method is suitable for the panel data that we have obtained, namely a composition of cross-section data represented by different countries and time series data in the form of yearly collected data points.

As this paper is using non-experimental data without separating a control and treatment group and no proper randomization can be ensured, we cannot draw causal inferences from the results. By adding control variables, the omitted variable bias is reduced. Reducing omitted variable bias helps in getting more accurate results regarding the association between poverty and out-of-pocket health expenditure attempting to approximate the intended causal effect. Heteroskedasticity robust standard errors are employed, allowing for both period and cross-section clustering.

The fixed effects model using linear regression has the following general form:

$$y_{it} = X_{it}\,\beta + a_i + c_t + \varepsilon_{it}$$

In the equation presented above, i represents a country and t represents a year. y_{it} accordingly is a data point of one of the four poverty measures in country i at time t. Since the data form an unbalanced panel, not all values for all combinations of i and t are available. X_{it} represents the data collected for a specific country i at time t and β indicates slope parameters for the different independent variables. The vector a_i consists only of one value across the whole vector which represents a cross-section-specific intercept, in our case a country-specific one. The vector c_t consists of only one value across the whole vector and represents a time series-specific intercept, in our case of a year. Finally, ε_{it} is the error term of each observation in country i at time t.

In our research, we employ the following model following the above given specification:

$$\text{Poverty headcount ratio at 2.15 USD} = b_0 + b_1 \cdot \text{out-of-pocket health expenditure as share of total health expenditure} + b_2 \cdot \log(\text{real GDP per capita}) + b_3 \cdot \text{unemployment rate} + b_4 \cdot \text{rural population as share of total population} + b_5 \cdot \text{female participation as share of total workforce} + a + c$$

We also estimate the model using other dependent variables. To increase robustness, we use the poverty gap, the poverty gap squared and the Watts index besides the poverty headcount ratio. Additionally, the model is estimated using different poverty lines to see whether the results are more generally applicable. Furthermore, the model is estimated separating the dataset into income groups following the classification of countries by the World Bank (2023).

Basically, we are testing the following hypotheses with our model:

$$H_0 : b_1 = 0 \text{ versus } H_1 : b_1 \neq 0$$

H_0 : out-of-pocket health expenditures as share of total health expenditures has no effect on the poverty headcount ratio (or other poverty indicators), keeping everything else constant.
H_1 : out-of-pocket health expenditures as share of total health expenditures has an effect on the poverty headcount ratio (or other poverty indicators), keeping everything else constant.

Results

In this section, we present and interpret the results of our regression analysis. The first regression was run using only the main variable of interest as the single regressor. The results that we have obtained are presented in Table 4.

Table 4 – Single OLS fixed effects regression results

	Poverty headcount ratio 2.15 USD	Poverty headcount ratio 3.65 USD	Poverty headcount ratio 6.85 USD
Share of out-of-pocket expenditure	0.1673***	0.2207**	0.1635
	(0.0450)	(0.0797)	(0.1324)
Constant	2.1390	8.4804***	24.3905
	(1.4192)	(2.5034)	(4.1565)

*Notes: this table reports the results of the OLS regression with fixed effects using only the main variable of interest out-of-pocket health expenditures as share of total health expenditures. The dependent variables are the poverty headcount ratio using the three different poverty lines. Standard errors are reported in parenthesis. Significance stars are reported based on the p-value of each estimated coefficient where * indicates p < 0.10, ** indicates p < 0.05 and *** indicates p < 0.01.*

Considering Table 4, we can conclude that there is a statistically significant positive association between the share of out-of-pocket expenditures on healthcare and poverty when looking at the poverty lines of 2.15 USD per day and 3.65 USD per day. The results in Tables D1 and D2 in Appendix D further substantiate these findings as the results of all four proxies for poverty have significant effects. The association between out-of-pocket health expenditures and the poverty headcount ratio employing the poverty line of 6.85 USD per day is not significant. On the other hand, the association between out-of-pocket health expenditures and the poverty gap, poverty gap squared and Watts index is significant and positive as shown in Table D3 in Appendix D. Considering the fact that the F-test of the model using all four proxies for poverty at the three poverty lines has a p-value of 0.0000, we can conclude that all of the employed regression models are statistically significant.

However, as mentioned before, we cannot directly infer causality from this model. In order to minimize the omitted variable bias in our initial model, we add the control variables that we have labeled as relevant in the theoretical framework section.

Table 5 – Multivariate OLS fixed effects regression results

	Poverty headcount ratio 2.15 USD	Poverty headcount ratio 3.65 USD	Poverty headcount ratio 6.85 USD
Share of out-of-pocket expenditure	0.1138**	0.1501**	0.0941
	(0.0459)	(0.0615)	(0.1156)
GDP per Capita	-12.2025***	-30.2822***	-42.3364***
	(2.5005)	(3.9681)	(5.6870)
Unemployment Rate	-0.1775*	-0.3159*	0.02739
	(0.0978)	(0.1703)	(0.17911)
Share of Rural Population	0.4533**	0.4144	0.2328
	(0.1912)	(0.2659)	(0.2802)
Share of Female Workforce Participation	0.6375*	0.7742	0.6879
	(0.3094)	(0.4484)	(0.4426)
Constant	79.6740**	258.2462***	398.3410***
	(31.0751)	(42.0920)	(58.8225)

*Notes: this table reports the results of the OLS regression with fixed effects using the main variable of interest out-of-pocket health expenditures as share of total health expenditures and all control variables. The dependent variables are the poverty headcount ratio using the three different poverty lines. Standard errors are reported in parenthesis. Significance stars are reported based on the p-value of each estimated coefficient where * indicates p < 0.10, ** indicates p < 0.05 and *** indicates p < 0.01.*

Table 5 shows the results of the multivariate OLS regression using fixed effects. It becomes clear that there still is a statistically significant positive association between the share of out-of-pocket expenditures on healthcare and poverty when looking at the poverty lines of 2.15 USD per day and 3.65 USD per day. We can see that after adding all the control variables to the model, the coefficient of our main variable has a smaller absolute value in all model specifications compared to the single regression model presented in Table 4. This indicates that we were able to reduce some of the omitted variable bias.

The second column of Table 5 presents the multivariate regression results when considering the poverty line of 2.15 USD per day. The main independent variable of interest, the out-of-pocket health expenditures as a share of total health expenditures, is significantly and positively associated with the poverty headcount ratio. Additionally, the results from Table E1 in Appendix E substantiate these results by looking at the other proxies for poverty, which show a significant positive association as well. All of the added control variables are significant. The natural logarithm of real GDP per capita and the poverty headcount ratio are

significantly and positively associated. Unemployment is negatively associated with the poverty headcount ratio. However, it is significant only at a ten percent significance level. The rural population as a share of the total population is significantly and positively associated with the poverty headcount ratio. The same association is apparent between the share of females as a percentage of the total labor force and the poverty headcount ratio.

In column three of Table 5, the same multivariate regression is run as in the second column. The only difference is that the poverty headcount ratio is now computed using the poverty line of 3.65 USD per day instead of 2.15 USD. Out-of-pocket health expenditures as a share of total health expenditures are again significantly and positively associated with the poverty headcount ratio. Table E2 in Appendix E substantiates these results by looking at the other proxies for poverty, which show a significant positive association as well. In this regression, the rural population as a share of the total population and the share of females as a percentage of the total labor force are not significant. The natural logarithm of GDP per capita is again significantly positively associated with the poverty headcount ratio. Unemployment is negatively associated with the poverty headcount ratio.

The last column of Table 5 shows the same multivariate regression as both prior discussed columns with the difference that the poverty line of 6.85 USD per day is now used to compute the poverty headcount ratio. Out-of-pocket health expenditures as a share of total health expenditures are not significantly associated with poverty when considering the poverty headcount ratio. In Table E3 in Appendix E, it can be seen that the other proxies for poverty are significantly positively associated with out-of-pocket health expenditures. The unemployment rate, the rural population as a share of the total population and the share of females as a percentage of the total labor force is not significant in the regression in the last column of Table 5. Real GDP per capita is significantly negatively associated with the poverty headcount ratio.

Tables F1, F2 and F3 in Appendix F present the regression results in the different income groups as classified by the World Bank (2023). It is to note that a regression for the low-income group is not included because too few observations from that group are present in the data sample to run a regression using fixed effects. Taking the results from Table F2 and F3 into account, we conclude that there is a significant positive association between the poverty headcount ratio using the extreme poverty line of 2.15 USD per day. In the high-income group, this association is not significant.

We denote that the overall significance of our model specifications in Table 5 is significant as the F-test results for the models yielded a p-value of 0.0000. Therefore, we have significant evidence to reject the null hypothesis and to conclude that the coefficient of out-of-pocket healthcare expenditures on poverty measures is different from 0 when the poverty lines of 2.15 USD and 3.65 USD per day are used. The results when the poverty line is 6.85 USD per day are not unitarily significant.

Conclusion and discussion

This research aims to identify the effect of out-of-pocket health expenditures as a share of total health expenditure on poverty. It uses four different proxies and three different lines for poverty and covers 143 countries during the period from 2000 to 2019. Considering the empirical analysis of this relation in this paper, it can be concluded that out-of-pocket health expenditures as a share of total health expenditure have a significant positive association with poverty across considering the poverty lines of 2.15 USD and 3.65 USD. Therefore, the null hypothesis is rejected. It is not possible to infer causal effects from our regression analysis as it is not properly randomized and there possibly exist other unobserved factors which relate to the dependent variable and main variable of interest. Given the sample that we use and keeping everything else constant, Table 5 shows that an increase in the out-of-pocket health expenditures as a share of total health expenditures of one percent leads to, on average, a 0.1138 (0.1501) percent increase in the poverty headcount ratio when employing a poverty line of 2.15 (3.65) USD per day. The result is statistically significant at the one percent confidence level.

The obtained results add to the existing academic literature by Sirag and Nor (2021) who used a dynamic panel threshold regression on worldwide data. They concluded that for lower-income countries, out-of-pocket health expenditures as a share of total health expenditures led to higher poverty rates. While their study used an older poverty line and included only real GDP per capita, our research continues to investigate this relation using altered poverty lines and more control variables.

Apart from the non-experimental nature of the data that is used in this paper, there are some other limitations to this study. Data used in this study has been collected from 143

countries but the number of observations differs per country and it is assumed that the remaining countries for which no complete observations exist are not random. It has already been hinted that data is missing primarily from countries in lower-income regions leading to bias in the results. Furthermore, the data regarding the poverty headcount ratio, the poverty gap and the poverty gap squared have been measured using income in some countries and consumption in other countries. These two measures can differ, especially in high-income countries. Additional research can be performed when more complete data is available to reduce bias and further study income and regional differences of the effects of out-of-pocket health expenditure on poverty on an aggregate level.

As our results showed differences in significance when using different poverty lines and poverty measures, follow-up research would be interesting to gain more insights into the different underlying mechanisms. For example, as the poverty headcount ratio and the poverty gap showed differences in various regressions in our analysis, further research can focus on disaggregating the effect of certain variables on poverty. The results of such research also are of high interest for policy implications if the effects of certain variables on different measures of poverty become clearer.

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Appendix A – Countries used in the regression analysis

Table A1 – Countries used in the regression analysis

country name	ISO 3166-1 alpha-3 codes	region	income group
Angola	AGO	Sub-Saharan Africa	Lower middle income
Albania	ALB	Europe & Central Asia	Upper middle income
United Arab Emirates	ARE	Middle East & North Africa	High income
Argentina	ARG	Latin America & Caribbean	Upper middle income
Armenia	ARM	Europe & Central Asia	Upper middle income
Australia	AUS	East Asia & Pacific	High income
Austria	AUT	Europe & Central Asia	High income
Azerbaijan	AZE	Europe & Central Asia	Upper middle income
Burundi	BDI	Sub-Saharan Africa	Low income
Belgium	BEL	Europe & Central Asia	High income
Benin	BEN	Sub-Saharan Africa	Lower middle income
Burkina Faso	BFA	Sub-Saharan Africa	Low income
Bangladesh	BGD	South Asia	Lower middle income
Bulgaria	BGR	Europe & Central Asia	Upper middle income
Bosnia and Herzegovina	BIH	Europe & Central Asia	Upper middle income
Belarus	BLR	Europe & Central Asia	Upper middle income
Bolivia	BOL	Latin America & Caribbean	Lower middle income
Brazil	BRA	Latin America & Caribbean	Upper middle income
Bhutan	BTN	South Asia	Lower middle income
Botswana	BWA	Sub-Saharan Africa	Upper middle income
Canada	CAN	North America	High income
Switzerland	CHE	Europe & Central Asia	High income
Chile	CHL	Latin America & Caribbean	High income
China	CHN	East Asia & Pacific	Upper middle income
Cote d'Ivoire	CIV	Sub-Saharan Africa	Lower middle income
Cameroon	CMR	Sub-Saharan Africa	Lower middle income
Democratic Republic of the Congo	COD	Sub-Saharan Africa	Low income
Republic of the Congo	COG	Sub-Saharan Africa	Lower middle income
Colombia	COL	Latin America & Caribbean	Upper middle income
Comoros	COM	Sub-Saharan Africa	Lower middle income
Cape Verde	CPV	Sub-Saharan Africa	Lower middle income

Costa Rica	CRI	Latin America & Caribbean	Upper middle income
Cyprus	CYP	Europe & Central Asia	High income
Czech Republic	CZE	Europe & Central Asia	High income
Germany	DEU	Europe & Central Asia	High income
Djibouti	DJI	Middle East & North Africa	Lower middle income
Denmark	DNK	Europe & Central Asia	High income
Dominican Republic	DOM	Latin America & Caribbean	Upper middle income
Ecuador	ECU	Latin America & Caribbean	Upper middle income
Spain	ESP	Europe & Central Asia	High income
Estonia	EST	Europe & Central Asia	High income
Ethiopia	ETH	Sub-Saharan Africa	Low income
Finland	FIN	Europe & Central Asia	High income
Fiji	FJI	East Asia & Pacific	Upper middle income
France	FRA	Europe & Central Asia	High income
Gabon	GAB	Sub-Saharan Africa	Upper middle income
United Kingdom	GBR	Europe & Central Asia	High income
Georgia	GEO	Europe & Central Asia	Upper middle income
Ghana	GHA	Sub-Saharan Africa	Lower middle income
Guinea	GIN	Sub-Saharan Africa	Low income
The Gambia	GMB	Sub-Saharan Africa	Low income
Guinea-Bissau	GNB	Sub-Saharan Africa	Low income
Greece	GRC	Europe & Central Asia	High income
Guatemala	GTM	Latin America & Caribbean	Upper middle income
Honduras	HND	Latin America & Caribbean	Lower middle income
Croatia	HRV	Europe & Central Asia	High income
Haiti	HTI	Latin America & Caribbean	Lower middle income
Hungary	HUN	Europe & Central Asia	High income
Indonesia	IDN	East Asia & Pacific	Lower middle income
India	IND	South Asia	Lower middle income
Ireland	IRL	Europe & Central Asia	High income
Iran	IRN	Middle East & North Africa	Lower middle income
Iraq	IRQ	Middle East & North Africa	Upper middle income
Iceland	ISL	Europe & Central Asia	High income
Israel	ISR	Middle East & North Africa	High income
Italy	ITA	Europe & Central Asia	High income
Jamaica	JAM	Latin America & Caribbean	Upper middle income
Jordan	JOR	Middle East & North Africa	Upper middle income
Japan	JPN	East Asia & Pacific	High income

Kazakhstan	KAZ	Europe & Central Asia	Upper middle income
Kenya	KEN	Sub-Saharan Africa	Lower middle income
Kyrgyzstan	KGZ	Europe & Central Asia	Lower middle income
Laos	LAO	East Asia & Pacific	Lower middle income
Liberia	LBR	Sub-Saharan Africa	Low income
Sri Lanka	LKA	South Asia	Lower middle income
Lesotho	LSO	Sub-Saharan Africa	Lower middle income
Lithuania	LTU	Europe & Central Asia	High income
Luxembourg	LUX	Europe & Central Asia	High income
Latvia	LVA	Europe & Central Asia	High income
Morocco	MAR	Middle East & North Africa	Lower middle income
Moldova	MDA	Europe & Central Asia	Upper middle income
Madagascar	MDG	Sub-Saharan Africa	Low income
Maldives	MDV	South Asia	Upper middle income
Mexico	MEX	Latin America & Caribbean	Upper middle income
North Macedonia	MKD	Europe & Central Asia	Upper middle income
Mali	MLI	Sub-Saharan Africa	Low income
Malta	MLT	Middle East & North Africa	High income
Myanmar	MMR	East Asia & Pacific	Lower middle income
Montenegro	MNE	Europe & Central Asia	Upper middle income
Mongolia	MNG	East Asia & Pacific	Lower middle income
Mozambique	MOZ	Sub-Saharan Africa	Low income
Mauritania	MRT	Sub-Saharan Africa	Lower middle income
Mauritius	MUS	Sub-Saharan Africa	Upper middle income
Malawi	MWI	Sub-Saharan Africa	Low income
Malaysia	MYS	East Asia & Pacific	Upper middle income
Namibia	NAM	Sub-Saharan Africa	Upper middle income
Niger	NER	Sub-Saharan Africa	Low income
Nigeria	NGA	Sub-Saharan Africa	Lower middle income
Nicaragua	NIC	Latin America & Caribbean	Lower middle income
Netherlands	NLD	Europe & Central Asia	High income
Norway	NOR	Europe & Central Asia	High income
Nepal	NPL	South Asia	Lower middle income
Pakistan	PAK	South Asia	Lower middle income
Panama	PAN	Latin America & Caribbean	High income
Peru	PER	Latin America & Caribbean	Upper middle income
Philippines	PHL	East Asia & Pacific	Lower middle income
Poland	POL	Europe & Central Asia	High income

Portugal	PRT	Europe & Central Asia	High income
Paraguay	PRY	Latin America & Caribbean	Upper middle income
Romania	ROU	Europe & Central Asia	High income
Russia	RUS	Europe & Central Asia	Upper middle income
Rwanda	RWA	Sub-Saharan Africa	Low income
Sudan	SDN	Sub-Saharan Africa	Low income
Senegal	SEN	Sub-Saharan Africa	Lower middle income
Solomon Islands	SLB	East Asia & Pacific	Lower middle income
Sierra Leone	SLE	Sub-Saharan Africa	Low income
El Salvador	SLV	Latin America & Caribbean	Lower middle income
Serbia	SRB	Europe & Central Asia	Upper middle income
Sao Tome and Principe	STP	Sub-Saharan Africa	Lower middle income
Slovakia	SVK	Europe & Central Asia	High income
Slovenia	SVN	Europe & Central Asia	High income
Sweden	SWE	Europe & Central Asia	High income
Eswatini	SWZ	Sub-Saharan Africa	Lower middle income
Chad	TCD	Sub-Saharan Africa	Low income
Togo	TGO	Sub-Saharan Africa	Low income
Thailand	THA	East Asia & Pacific	Upper middle income
Tajikistan	TJK	Europe & Central Asia	Lower middle income
East Timor	TLS	East Asia & Pacific	Lower middle income
Tonga	TON	East Asia & Pacific	Upper middle income
Tunisia	TUN	Middle East & North Africa	Lower middle income
Turkey	TUR	Europe & Central Asia	Upper middle income
Tanzania	TZA	Sub-Saharan Africa	Lower middle income
Uganda	UGA	Sub-Saharan Africa	Low income
Ukraine	UKR	Europe & Central Asia	Lower middle income
Uruguay	URY	Latin America & Caribbean	High income
United States of America	USA	North America	High income
Uzbekistan	UZB	Europe & Central Asia	Lower middle income
Vietnam	VNM	East Asia & Pacific	Lower middle income
Vanuatu	VUT	East Asia & Pacific	Lower middle income
Samoa	WSM	East Asia & Pacific	Lower middle income
South Africa	ZAF	Sub-Saharan Africa	Upper middle income
Zambia	ZMB	Sub-Saharan Africa	Low income
Zimbabwe	ZWE	Sub-Saharan Africa	Lower middle income

Notes: This table shows the countries used in the regression analysis. For each country, the ISO 3166-1 alpha codes and the region and income group according to the World Bank are included.

Appendix B – Descriptive statistics of the poverty indicators

Table B1 – Descriptive statistics of the poverty indicators

Variable	Obs.	Mean	St. Dev.	Min.	Max.
Poverty headcount ratio 2.15 USD	1449	7.41	14.77	0.00	91.47
Poverty headcount ratio 3.65 USD	1449	15.43	23.09	0.00	97.57
Poverty headcount ratio 6.85 USD	1449	29.54	31.39	0.00	99.70
Poverty gap 2.15 USD	1449	2.68	6.22	0.00	58.49
Poverty gap 3.65 USD	1449	6.29	11.29	0.00	73.61
Poverty gap 6.85 USD	1449	14.01	18.54	0.00	85.49
Poverty gap squared 2.15 USD	1449	1.40	3.54	0.00	41.47
Poverty gap squared 3.65 USD	1449	3.49	7.07	0.00	58.64
Poverty gap squared 6.85 USD	1449	8.59	13.05	0.00	74.84
Watts index 2.15 USD	1449	3.90	9.84	0.00	89.61
Watts index 3.65 USD	1449	9.74	19.14	0.00	89.91
Watts index 6.85 USD	1449	23.59	34.97	0.00	92.20

Notes: This table shows additional descriptive statistics for all poverty measures using the three different poverty lines as specified by the World Bank. The number of observations per variable in the sample, as well as the sample mean, sample standard deviation, minimum and maximum values are recorded. All units are measured in accordance with the units presented in Table 1.

Appendix C – Scatterplots of different income groups

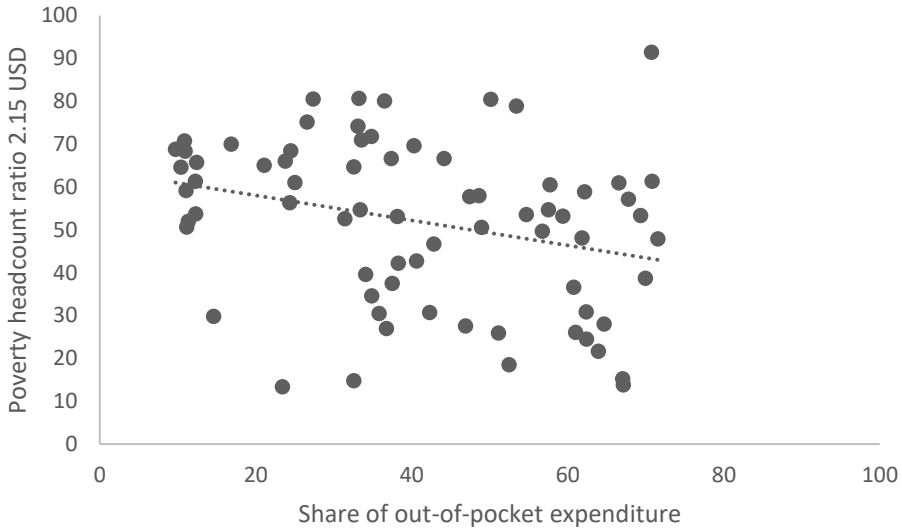

Figure C1 – Scatterplot showing low income countries

Notes: The scatterplot shows the relationship between the out-of-pocket health expenditures as share of total health expenditures and the poverty headcount ratio using a poverty line of 2.15 USD per day in low income countries. Both measurements are given as percentages between 0 and 100. Each point on the graph represents one observation of the dataset.

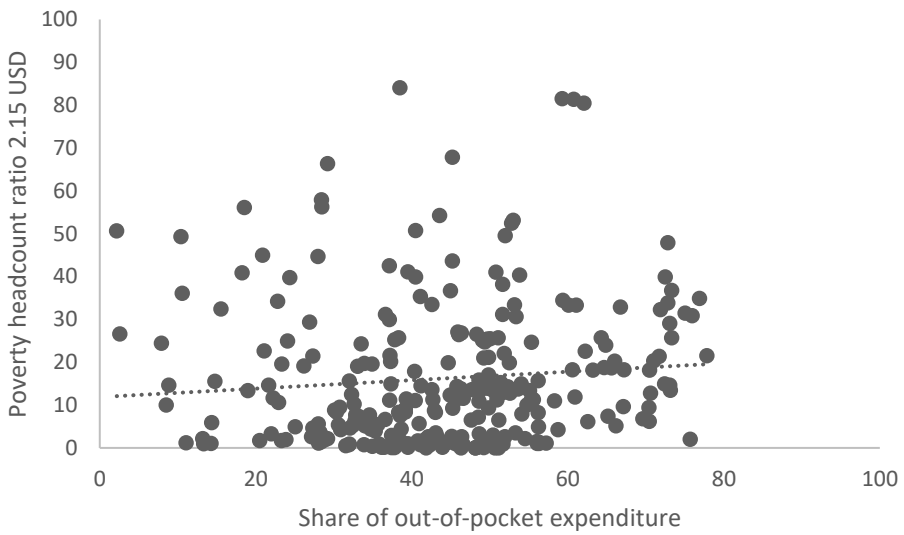

Figure C2 – Scatterplot showing lower-middle income countries

Notes: The scatterplot shows the relationship between the out-of-pocket health expenditures as share of total health expenditures and the poverty headcount ratio using a poverty line of 2.15 USD per day in lower-middle income countries. Both measurements are given as percentages between 0 and 100. Each point on the graph represents one observation of the dataset.

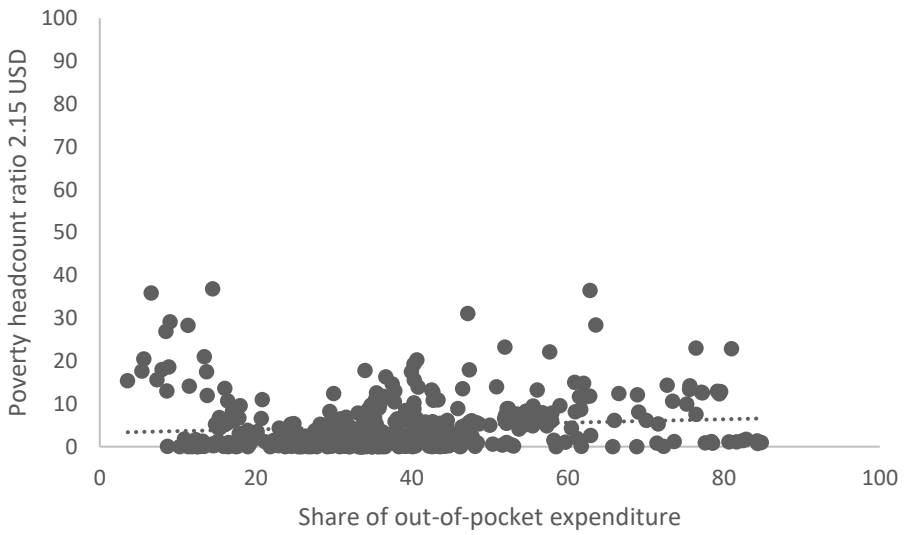

Figure C3 – Scatterplot showing upper-middle income countries

Notes: The scatterplot shows the relationship between the out-of-pocket health expenditures as share of total health expenditures and the poverty headcount ratio using a poverty line of 2.15 USD per day in upper-middle income countries. Both measurements are given as percentages between 0 and 100. Each point on the graph represents one observation of the dataset.

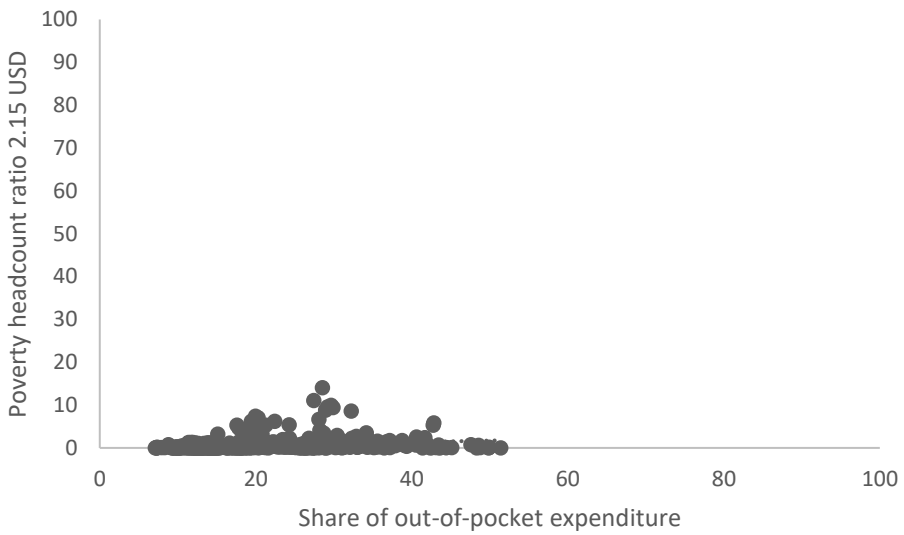

Figure C4 – High income countries

Notes: The scatterplot shows the relationship between the out-of-pocket health expenditures as share of total health expenditures and the poverty headcount ratio using a poverty line of 2.15 USD per day in high income countries. Both measurements are given as percentages between 0 and 100. Each point on the graph represents one observation of the dataset.

Appendix D – Single OLS fixed effects regression results

Table D1 – Single OLS fixed effects regression results for the 2.15 USD poverty line

	Poverty headcount ratio 2.15 USD	Poverty gap 2.15 USD	Poverty gap squared 2.15 USD	Watts index 2.15 USD
Share of out-of-pocket expenditure	0.1673***	0.0812***	0.0509***	0.1351***
	(0.0450)	(0.0246)	(0.0176)	(0.0447)
Constant	2.1390	0.1203	-0.2040	-0.3543
	(1.4192)	(0.7781)	(0.5534)	(1.4113)

*Notes: this table reports the results of the OLS regression with fixed effects using only the main variable of interest out-of-pocket health expenditures as share of total health expenditures. The dependent variables are the four indicators for poverty. The used poverty line is 2.15 USD per day. Standard errors are reported in parenthesis. Significance stars are reported based on the p-value of each estimated coefficient where * indicates p < 0.10, ** indicates p < 0.05 and *** indicates p < 0.01.*

Table D2 – Single OLS fixed effects regression results for the 3.65 USD poverty line

	Poverty headcount ratio 3.65 USD	Poverty gap 3.65 USD	Poverty gap squared 3.65 USD	Watts index 3.65 USD
Share of out-of-pocket expenditure	0.2207**	0.1298***	0.0892***	0.2385***
	(0.0797)	(0.0346)	(0.0244)	(0.0645)
Constant	8.4804***	2.2030*	0.6828	2.2287
	(2.5034)	(1.0905)	(0.7712)	(2.0328)

*Notes: this table reports the results of the OLS regression with fixed effects using only the main variable of interest out-of-pocket health expenditures as share of total health expenditures. The dependent variables are the four indicators for poverty. The used poverty line is 3.65 USD per day. Standard errors are reported in parenthesis. Significance stars are reported based on the p-value of each estimated coefficient where * indicates p < 0.10, ** indicates p < 0.05 and *** indicates p < 0.01.*

Table D3 – Single OLS fixed effects regression results for the 6.85 USD poverty line

	Poverty headcount ratio 6.85 USD	Poverty gap 6.85 USD	Poverty gap squared 6.85 USD	Watts index 6.85 USD
Share of out-of-pocket expenditure	0.1635	0.1666**	0.1377***	0.3698***
	(0.1324)	(0.0648)	(0.0415)	(0.1113)
Constant	24.3905	8.7654***	4.2541***	11.9394***
	(4.1565)	(2.0358)	(1.3055)	(3.5003)

*Notes: this table reports the results of the OLS regression with fixed effects using only the main variable of interest out-of-pocket health expenditures as share of total health expenditures. The dependent variables are the four indicators for poverty. The used poverty line is 6.85 USD per day. Standard errors are reported in parenthesis. Significance stars are reported based on the p-value of each estimated coefficient where * indicates p < 0.10, ** indicates p < 0.05 and *** indicates p < 0.01.*

Appendix E – Multivariate OLS fixed effects regression results

Table E1 – Multivariate OLS fixed effects regression results for the 2.15 USD poverty line

	Poverty headcount ratio 2.15 USD	Poverty gap 2.15 USD	Poverty severity index 2.15 USD	Watts index 2.15 USD
Share of out-of-pocket expenditure	0.1138**	0.0595**	0.0390**	0.1038**
	(0.0459)	(0.0258)	(0.0180)	(0.0462)
Real GDP per capita	-12.2025***	-3.1198**	-1.1908*	-4.3868**
	(2.5005)	(1.1272)	(0.6794)	(1.8594)
Unemployment rate	-0.1775*	-0.0459	-0.0141	-0.0999*
	(0.0978)	(0.0343)	(0.0197)	(0.0557)
Share of rural population	0.4533**	0.2066**	0.1193**	0.3051**
	(0.1912)	(0.0770)	(0.0448)	(0.1211)
Share of female workforce participation	0.6375*	0.2729**	0.14465**	0.4506**
	(0.3094)	(0.1179)	(0.0671)	(0.1825)
Constant	79.6740**	12.2081	1.3001	13.5319
	(31.0751)	(14.8213)	(8.8878)	(24.0703)

*Notes: this table reports the results of the OLS regression with fixed effects using only the main variable of interest out-of-pocket health expenditures as share of total health expenditures and all control variables. The dependent variables are the four indicators for poverty. The used poverty line is 2.15 USD per day. Standard errors are reported in parenthesis. Significance stars are reported based on the p-value of each estimated coefficient where * indicates p < 0.10, ** indicates p < 0.05 and *** indicates p < 0.01.*

Table E2 – Multivariate OLS fixed effects regression results for the 3.65 USD poverty line

	Poverty headcount ratio 3.65 USD	Poverty gap 3.65 USD	Poverty severity index 3.65 USD	Watts index 3.65 USD
Share of out-of-pocket expenditure	0.1501**	0.0908**	0.0643**	0.1743**
	(0.0615)	(0.0332)	(0.0252)	(0.0654)
Real GDP per capita	-30.2822***	-10.8209***	-5.0971***	-15.6278***
	(3.9681)	(1.8151)	(1.1861)	(3.1747)
Unemployment rate	-0.3159*	-0.1371*	-0.0684	-0.2491*
	(0.1703)	(0.0756)	(0.0428)	(0.1195)
Share of rural population	0.4144	0.3057**	0.2185**	0.5372**
	(0.2659)	(0.1343)	(0.0850)	(0.2250)
Share of female workforce participation	0.7742	0.4616**	0.3027**	0.8441**
	(0.4484)	(0.2201)	(0.1336)	(0.3520)
Constant	258.2462***	78.4511***	30.4914*	101.9121**
	(42.0920)	(21.5496)	(15.0601)	(39.1136)

*Notes: this table reports the results of the OLS regression with fixed effects using only the main variable of interest out-of-pocket health expenditures as share of total health expenditures and all control variables. The dependent variables are the four indicators for poverty. The used poverty line is 3.65 USD per day. Standard errors are reported in parenthesis. Significance stars are reported based on the p-value of each estimated coefficient where * indicates p < 0.10, ** indicates p < 0.05 and *** indicates p < 0.01.*

Table E3 – Multivariate OLS fixed effects regression results for the 6.85 USD poverty line

	Poverty headcount ratio 6.85 USD	Poverty gap 6.85 USD	Poverty severity index 6.85 USD	Watts index 6.85 USD
Share of out-of-pocket expenditure	0.0941	0.1120**	0.0953**	0.2609**
	(0.1156)	(0.0534)	(0.0362)	(0.0990)
Real GDP per capita	-42.3364***	-23.8756***	-14.8769***	-39.7142***
	(5.6870)	(3.0497)	(2.0735)	(5.5370)
Unemployment rate	0.02739	-0.1547	-0.1388	-0.3828*
	(0.17911)	(0.1160)	(0.0847)	(0.2117)
Share of rural population	0.2328	0.3110	0.2902**	0.7338*
	(0.2802)	(0.1816)	(0.1385)	(0.3514)
Share of female workforce participation	0.6879	0.5874**	0.4783**	1.3164**
	(0.4426)	(0.2668)	(0.2146)	(0.5185)
Constant	398.3410***	206.4579***	119.7390***	319.9355***
	(58.8225)	(31.8639)	(22.8687)	(60.7263)

*Notes: this table reports the results of the OLS regression with fixed effects using only the main variable of interest out-of-pocket health expenditures as share of total health expenditures and all control variables. The dependent variables are the four indicators for poverty. The used poverty line is 6.85 USD per day. Standard errors are reported in parenthesis. Significance stars are reported based on the p-value of each estimated coefficient where * indicates p < 0.10, ** indicates p < 0.05 and *** indicates p < 0.01.*

Appendix F – Multivariate OLS fixed effects regression results for different income groups

Table F1 – Multivariate OLS fixed effects regression results for the high income group

	Poverty headcount ratio 2.15 USD	Poverty headcount ratio 3.65 USD	Poverty headcount ratio 6.85 USD
Share of out-of-pocket expenditure	0.0140	0.0268	-0.0005
	(0.0273)	(0.0580)	(0.1288)
Real GDP per capita	-5.1653*	-14.7170***	-42.0116***
	(2.6726)	(3.7472)	(7.7888)
Unemployment rate	-0.0363	-0.1673*	-0.4410*
	(0.0366)	(0.0874)	(0.2363)
Share of rural population	0.1505	0.1855	-0.0363
	(0.1158)	(0.1510)	(0.2505)
Share of female workforce participation	-0.1825	-0.0017	0.7888
	(0.1113)	(0.2795)	(0.7700)
Constant	59.5454*	152.7656***	415.9559***
	(29.5289)	(39.4655)	(68.7461)

*Notes: this table reports the results of the OLS regression with fixed effects using only the main variable of interest out-of-pocket health expenditures as share of total health expenditures and all control variables for all high income countries as specified by the World Bank. The dependent variables are the poverty headcount ratios at each of the three poverty lines. Standard errors are reported in parenthesis. Significance stars are reported based on the p-value of each estimated coefficient where * indicates p < 0.10, ** indicates p < 0.05 and *** indicates p < 0.01.*

Table F2 – Multivariate OLS fixed effects regression results for the upper-middle income group

	Poverty headcount ratio 2.15 USD	Poverty headcount ratio 3.65 USD	Poverty headcount ratio 6.85 USD
Share of out-of-pocket expenditure	0.1803***	0.2993***	0.2108
	(0.0421)	(0.0818)	(0.1863)
Real GDP per capita	-15.1165***	-36.8479***	-43.5322***
	(3.8046)	(6.5792)	(10.9377)
Unemployment rate	0.0235	0.0673	0.6135
	(0.1172)	(0.2771)	(0.4729)
Share of rural population	-0.1307	-0.4504	-0.2047
	(0.2101)	(0.3072)	(0.4145)
Share of female workforce participation	-0.1322	0.0994	0.9970
	(0.2380)	(0.4926)	(0.8254)
Constant	151.3056***	362.1673***	398.2931***
	(34.3964)	(59.8094)	(106.5240)

*Notes: this table reports the results of the OLS regression with fixed effects using only the main variable of interest out-of-pocket health expenditures as share of total health expenditures and all control variables for all upper-middle income countries as specified by the World Bank. The dependent variables are the poverty headcount ratios at each of the three poverty lines. Standard errors are reported in parenthesis. Significance stars are reported based on the p-value of each estimated coefficient where * indicates p < 0.10, ** indicates p < 0.05 and *** indicates p < 0.01.*

Table F3 – Multivariate OLS fixed effects regression results for the lower-middle income group

	Poverty headcount ratio 2.15 USD	Poverty headcount ratio 3.65 USD	Poverty headcount ratio 6.85 USD
Share of out-of-pocket expenditure	0.1651**	0.1182	-0.1339
	(0.0785)	(0.1193)	(0.1491)
Real GDP per capita	-21.5012***	-39.9258***	-24.6812**
	(6.4878)	(7.4043)	(9.1254)
Unemployment rate	0.4243	0.6676	1.1469**
	(0.6628)	(0.6748)	(0.4319)
Share of rural population	-0.1983	-0.6154	-0.1517
	(0.3829)	(0.6609)	(0.4119)
Share of female workforce participation	0.1877	0.3095	0.4243
	(0.4419)	(0.7127)	(0.4107)
Constant	194.4982***	390.9789***	266.4965***
	(64.3894)	(70.4083)	(72.6418)

*Notes: this table reports the results of the OLS regression with fixed effects using only the main variable of interest out-of-pocket health expenditures as share of total health expenditures and all control variables for all lower-middle income countries as specified by the World Bank. The dependent variables are the poverty headcount ratios at each of the three poverty lines. Standard errors are reported in parenthesis. Significance stars are reported based on the p-value of each estimated coefficient where * indicates p < 0.10, ** indicates p < 0.05 and *** indicates p < 0.01.*